A Fall Detection and Alert System for an Elderly using Computer Vision and Internet of Things

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Abstract —**Fall related injuries are major problems for elderly people leaving alone at home. Most of people who have fallen cannot get up without assistance. Absence of movement of person after fall may cause severe complications regarding health. Considering these problems for elderly we propose a fall detection system for elderly based on computer vision technology. Considering advantages of this technology over wearable based sensor such patients don't need to wear any sensor on body avoiding discomfort due to wearable one. We propose a system in which it is able to detect fall event by detecting features from scene captured by single camera. Features detected are orientation angle, aspect ratio, centre of mass and Hu moment invariants. To reduce computation time in real situation minimum features are used to detect fall. We have also applied IOT (Internet of Things) approach to notify about accident through email with attached screenshot of scene and video recorded after fall. One can also view live streaming of room by entering IP address of raspberry pi on browser if he wishes to see. Camera interfacing and internet connection is done through raspberry pi3 model B which is just like a mini computer with Operating system embedded into its SD card. Overall we propose an automated fall detection system with computer vision and IOT approach.**

Keywords — computer vision, internet of things, OPEN CV, python, raspberry pi.

I. INTRODUCTION

Now days due to busy schedule it's not always possible to keep someone at home to take care of elder person. Caretaker is the option for that but then also one cannot completely trust on caretaker by taking risk as that person is close to us and we are concerned about his or her health a lot. By 2050, 69% of all persons aged 80 or over are expected to be present in developing countries in future [1]. Most elder persons are living alone and according to [2] over 62% of the elderly patients were injured due to fall. In paper [3] it is proved that 5% to 10% of falls cause injuries such as fractures, head injury and from these accidents 60% are indoor accidents. Falls also lead to decreased independence and many people who have fallen are afraid of falling again. Tissue injuries, joint dislocations, bone fractures and head trauma are some of the damages caused by falling. Most people are unable to get up by themselves after fall and even if those who are not injured, half of those who had experienced lying on floor for much time generally more than an hour died within 6 months after that incident [4].

As it is essential to have a system which can detect fall instantly so that proper treatment at exact time can be given to patients, many techniques are evolved. Fall detection system can be of wearable sensors or external sensors. External sensors are placed around subject whereas wearable sensors are attached to subject's body.

Fall detection can be done by using camera effectively as it avoids the need for continuous attachment of sensors to body. Camera captures scene, foreground which is moving object in this case is detected and different features are extracted from that object. Extracted features are used to detect fall event.

This article presents algorithm to detect fall by analyzing video. Here we are using single camera to avoid need of calibrating cameras in multiple camera system. It reduces complexity and cost of the system. This system detects four movement features important to distinguish fall event from normal event. Aspect ratio, Orientation angle, center of mass, and Hu's moment invariants are these features which we detect from video. System involves 4 steps as: 1) detecting movement features 2) finding threshold for these features to detect fall 3) sending email along with saved screenshot and video after fall has been detected. In this system we have done a programming using python2.7 software along with OPEN CV 3.1. Person at remote side will be able to see the live streaming of room if he wishes to. Raspberry pi controller is used for camera interfacing and internet connection.

Here brief introduction about system has given in section I. Section II describes previous related work. Proposed method is presented in section III, remaining section IV gives results of experiments and discussion and conclusion of work done has been justified.

II. RELATED WORK

Variety of approaches have been taken to detect fall with varying degrees of accuracy .Most systems have used wearable sensor based and camera based approach in order to detect fall event.

In wearable sensor based approach many sensors like accelerometer, gyroscopes, heart rate sensors, magnetometer sensors or microphones are used. Widely used sensor is accelerometer .These sensors are used to access posture of body. These sensors are placed on body part of subject like waist, thigh, wrist, shoes etc. to detect fall. Whereas Camera based system captures the scene, detects the motion, extracts the movable object and then its features which are used to determine fall event. Mostly used approach for identifying the moving object is background subtraction, in which each frame is compared with background model. Pixels in current frame deviating from background are considered as moving

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object. This foreground region is used to extract features from it to detect fall. Features can be of many types like aspect ratio, distance of centroid of human body to the floor, fall angle, shape, center of mass, etc.

Frank Spasro et al. [5] developed android phone application to detect fall using an integrated accelerometers. Khali Nizmand et al. [6] introduced first ever washable garment to detect fall. Won-Jae Yi et al. [7] proposed a system collecting physiological data from body. Alert is through SMS [5] [7], internet [7]. In some sensor based system communication protocols such as zigbee, Bluetooth [8] [9] have been used to send alarm signals after fall event. From studies we can conclude that in all sensor based systems, overall common problems are:

1) It demands recharging of battery time to time.
2) Leads to discomfort due to continuous attachment of sensor.
3) System is expensive.
4) Most system assumes that patients are capable of pushing panic buttons and which is not possible always.

Considering these problems camera based approach comes into picture to avoid limitations regarding sensor based method. Vision based fall detectors are multi-camera based [11] [10] [12], depth camera based [13] [14], single camera based [15] [16] [17]. Mirmahboub et al. [15] extracted features from foreground region and used SVM classifier for classification. Rougier et al. [16] used shape matching algorithm to detect fall. Simple method was proposed in [18] which analyzed aspect ratio whereas method in [19] finds distance of people centroid from floor. In [20] falls have been identified by analyzing volume distribution along vertical axis.

To avoid cost of system, need of calibration of multiple cameras, need of video fusion hardware, need for synchronization of video sequence from each camera and difficult implementation of multiple camera system we have selected single camera approach so as to reduce cost and complexity of system in order to be affordable to all. Again although fall detection systems which can send alert when fall event occurs are available with different technologies, Internet of the Things approach to notify person at remote side is much better now days as this is the popular communication medium in today's world. Live streaming of area under camera after fall detection will ensure person at other side whether the patient is fine or not and according to that he can take quick decision. As Python OPEN CV is freely available, portable and does fast processing over Matlab, we preferred this platform. Raspberry pi has python language as its primary language, that's another reason we chose it.

III. PROPOSED METHOD

Briefly stated our method involves detecting movement and then extracting that movable object (In this case Patient), detecting four movement features from foreground object as aspect ratio, orientation angle, centroid and Hu moment invariants, taking decision whether fall has happened or not based on threshold found out during experimental analysis, and finally sending notification through email along with attached screenshot and video after fall has been detected. Person at remote side can see live streaming at any time if he wishes to see just by logging in IP address of raspberry pi. Programming is done in python language.

1 Foreground extraction

In this step we extract moving foreground region from static background. We are using Gaussian Mixture based foreground segmentation algorithm implemented in OPEN CV which is based on [21] [22]. This algorithm selects appropriate number of Gaussian distribution instead of fixed one for every pixel. It also has better adaptability to varying illumination. The processing time reduction and improved segmentation are important advantages of this method of background subtraction. In this case you can choose whether to detect shadows in scene or not. This is better model for complex dynamic scenes and also gives compact representation. After background estimation we determine foreground by simple background subtraction method in which difference between current frame and background frame is calculated. Let I(x, y, t) be the gray-level of a pixel by (x, y) coordinate in the frame at the time of t, t= 1..., n. where n is the total number of frames in a video sequence. The difference of frames can be defined by:

$$df(x,y,t) = \begin{cases} 1 & if\ I(x,y,t) - I(x,y,t-1) \geq Td \\ 0 & otherwise \end{cases} \tag{1}$$

Where, T_d is a threshold. If d_f(x, y, t) is equal to 1, this (x, y) is foreground. After foreground detection morphological processes such as opening and closing on foreground image have been used to get somewhat exact foreground image. Contour detection and drawing is the next step that we do on foreground image and thus identifying contour with maximum area as person in the room. This contour is further used for detection of movement features

2 Movement Feature Extraction

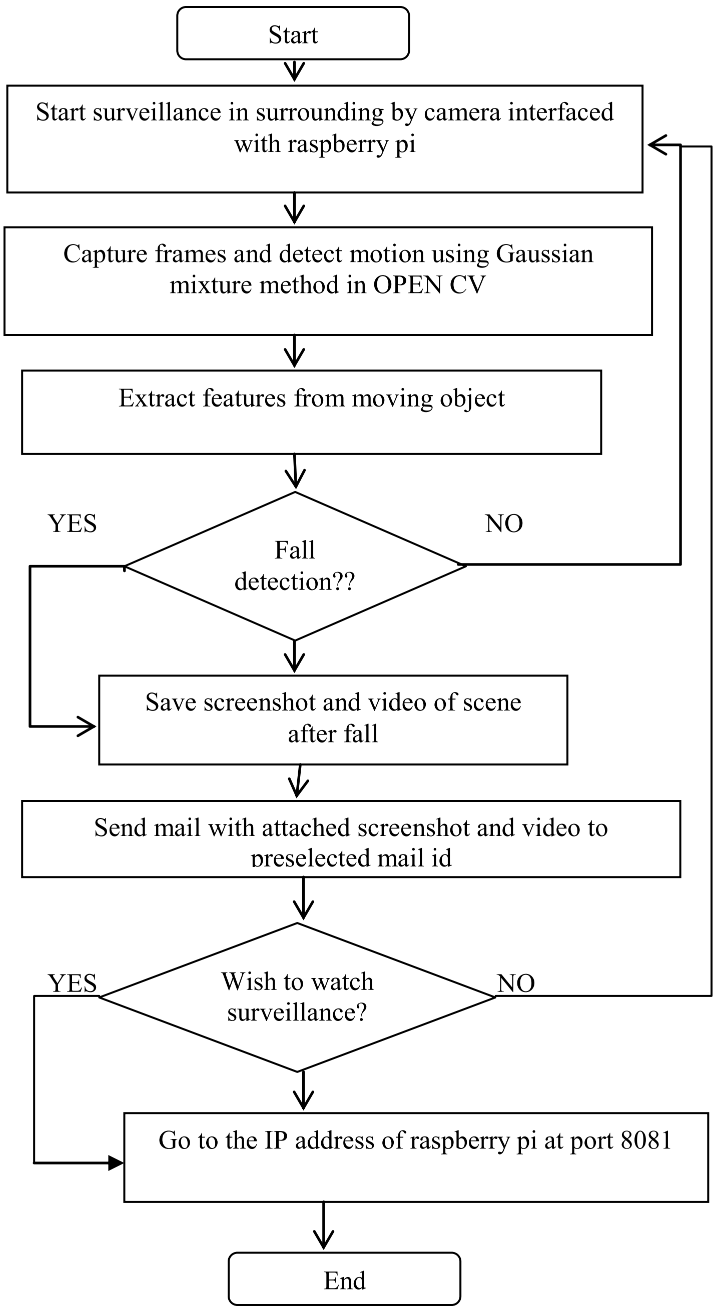
Using foreground that we detected, we now detect four features which can denote sudden change in movement. There are many features available but we decided to detect selected four features which can help to reduce computation and thus processing time of a system.

A) Aspect Ratio

Aspect ratio is the ratio of width of bounding box around foreground to its height described as below equation (2),

Aspect ratio= width/height (2)

Low aspect ratio denotes person is in standing position where as lying person has high aspect ratio. Here from observation of variation of aspect ratio of successive frames in video we set threshold for fall detection. Aspect ratio is greater than or equal to1.5 during fall.



Flowchart: proposed system

B) Center of Mass

It is a point where all mass of object is concentrated. Center of mass can be found by following equation (3),

$$Xcm,k = \frac{\sum_{i=1}^{n} Xi}{n},$$

$$Ycm,k = \frac{\sum_{i=1}^{n} yi}{n} \quad (3)$$

where n is the total number of foreground pixels of object K. and (x_i, y_i) is the coordinate of pixel i in it.We find change of center of mass during consecutive frames and by using predefined threshold obtained from observation we detect fall .

C) Orientation angle

This is the angle at which object is directed. It is angle between long axis of bounding ellipse and horizontal plane of image. Fall occurred when this angle is near to 90 degrees.

D) Change of Hu Moment Invariants

Image moment is weighted average of image pixel values while moment invariants are invariants that are formed from moments. The only moments that are invariants themselves are the central moments [23]. Widely used shape descriptor is Hu moment invariant, as they are independent to scaling, rotation and translation. Hu moments were introduced in 1960's by Hu [24]. These Hu moments are normally extracted from silhouette of an object which is foreground image in our case where Hu moments will be calculated over white pixels only. [25] 2-dimensional moments of order (p + q) of image that has grey function f (x, y) is defined as equation (4),

$$Mpq = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^p\, y^q f(x,y)\, dx\, dy \quad (4)$$

When a scaling normalization is applied the central moments are defined as (5),

$$u_{pq} = \sum_{x}\sum_{y} (x - x^{-})^p\, (y - y^{-})^q f(x,y) \quad (5)$$

Where f(x, y) is a digital image and $(\bar{x}, \bar{y})$ is the Center of mass as (6),

$$x^{-} = \frac{m_{10}}{m_{00}}, \; y^{-} = \frac{m_{01}}{m_{00}} \quad (6)$$

These central moments are translational invariants. Normalized central moments are scale invariant and calculated as per equation (7),

$$\eta_{ij} = \frac{u_{ij}}{u00^{(1+\frac{i+j}{2})}} \quad (7)$$

Where, $i+j \geq 2$.

Finally, discovered translational, scale and rotation invariant seven Hu moments discovered by Hu [26] [27] are presented as I_1 to I_7 as following equations (8),

$$I_1 = \eta_{20} + \eta_{02} \quad (8)$$

$$I_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$I_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2$$

$$I_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2$$

$$I_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

$$I_6 = (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03})$$

$$I_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] - (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

In order to compute these Hu Moments in OPEN CV platform, we first need to compute the original 24 moments associated with the image. From there, we pass these moments into cv2.HuMoments, which calculates Hu's seven invariant moments. Now, we get feature vector containing 7 values which changes according to change in shape of an object so by calculating distance between moment values for successive frames, we conclude that if change of values is more than 50% between successive frames then fall is detected.

3 Sending mail using SMTP

In order to send mail with attached screenshot and video of scene after fall event we use SMTP in python. Module named smtplib is useful for communicating with mail servers to send an email [28]. Steps to send email are as: 1) to create an SMTP object(one object for connection with one server), 2) log in to server, 3) convert file to be send into Base64 where Base64 is a encoding scheme that represent binary data in an ASCII string format. The term *Base64* originates from a specific MIME (Multipurpose Internet Mail Extensions) content transfer encoding. [29] 4) send the mail. (MIME) is an Internet standard that extends the format of email to support: [30]

- Text in character sets other than ASCII
- Non-text attachments: audio, video, images, application programs etc.
- Message bodies with multiple parts
- Header information in non-ASCII character sets Virtually all human-written Internet email and a fairly large proportion of automated email are transmitted via SMTP in MIME format.

4 Video surveillance availability

If person at remote side wish to watch live streaming of room through website then he can go to the IP address of raspberry pi on browser. In this we are using motion package available in raspbian OS. Motion is a program that monitors video signal from camera and is able to detect motion [31].By making some required changes in motion.conf file, we will be able to access webcam remotely and thus can see live video from any computer connected to local host. To access camera from all over in world we need to do port forwarding. Pi camera has better resolution but most USB cameras can be used for this purpose. We have also tried streaming with flask in python. Flask is a micro web application written in Python [32]. Flask provides support for streaming through the use of generator functions which can return multiple results in sequence and this is the property of generator function which is used by flask for live streaming. Streaming of independent JPEG pictures is the method that is used in flask streaming method and called as 'Motion JPEG'. This is used in many surveillance systems.

5 Hardware set up

In this project we are using Pi-camera for capturing scene as it has better resolution and compatibility with raspberry pi. We are using Raspberry Pi3 model B as it has on board Wi-Fi and Bluetooth. Here we are working with Raspbian OS with Python and OPEN CV embedded into it. Raspberry with camera connected is to be placed in room so that it can cover maximum area in room. Raspberry can be connected to router through Ethernet cable or Wi-Fi. On terminal of Raspbian, type "Python file path"and run the program and there system will start doing his work of detection of fall event. Interface with camera is as shown in figure 1. Check camera connection using lsusb command and Install fswebcam to work with camera.

Figure1. Interface of raspberry pi with camera

IV. RESULTS AND DISCUSSION

We have implemented our algorithm using OPENCV 3.1 and Python 2.7on windows 8.1 platform (64 bit operating system, Intel core, 2.6GHZ CPU and 4 GB RAM) at initial experiment stages. Database that we used is available at http://le2i.cnrs.fr/. Below we have given screenshots of results that we obtained during experiment such as Motion detection in fig.2, getting feature values from foreground and detecting fall are shown in fig3 and fig.4 respectively. After fall event has occurred screenshot of scene has been taken automatically by camera and saved, the same email has been sent to predefined mail id is shown in fig.5. Fig.6 is showing screenshot of live streaming on system connected to local network.

Dataset resource

We have worked initially on available database .Datasets of fall videos are obtained from http://le2i.cnrs.fr/. This is a dataset in realistic video surveillance setting obtained using a single camera. The frame rate is 25frames/s and the resolution is 320x240 pixels. The video data includes main difficulties that we can face in real time. Occlusion cluttered or textured background and varying illumination are problems encountered in video. Fall during various normal activities is the content of videos. This dataset contains 191 videos.

Figure2. Motion detection in a room.

Figure3. Feature detection from foreground.

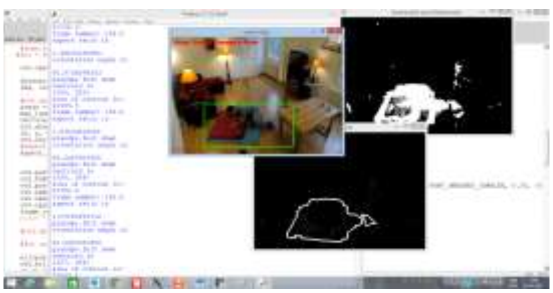


Figure4. Fall has been detected when feature values goes beyond threshold.

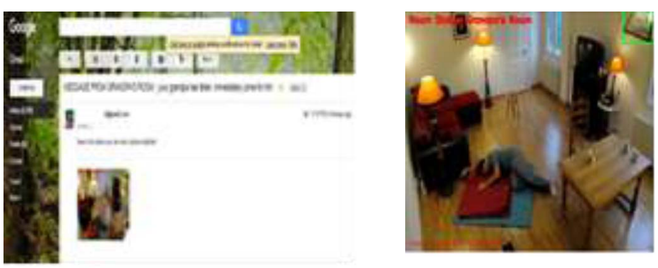


Figure5. Captured and saved screen shot after fall has been sent to email id

Figure6. Live streaming on computer connected to local

network using flask in python.

To describe our results we use three parameters with the following equations (9)

Capacity to detect fall, Sensitivity = TP/ (TP + FN).
Capacity to detect only a fall, Specificity= TN/ (TN+FP).
Ability to correctly detect fall and not fall event,
Accuracy= (TP +TN)/ (TN+TP=FP+FN). (9)

TABLE I. RESULTS OF PROPOSED METHOD AFTER ANALYSIS

Method	Selectivity (%)	Specificity (%)	Accuracy (%)
Proposed algorithm	88.23	92.98	91.89

Where TP, FN, FP and TN are respectively falls those are correctly detected, falls not detected, normal activities detected as a fall and normal activities not detected as a fall. According to results we got accuracy of near about 90% .By combining threshold for calculated four features We take a decision about fall happened or not. It exactly detects fall in many cases and program has execution time of near about 77.29 seconds for each video in dataset. Table I shows results of approach.

CONCLUSION AND FUTURE WORK

Using wearable sensors for fall detection is cost efficient, but leads to discomfort due to need for continuous attachment of sensors on body .Thus Vision based approach is better in this case where among various techniques for visual detection, single RGB camera approach is better, while considering cost of system and easy setup. In this paper we have reviewed various approaches taken to detect and notify fall event up till now. Using more features makes system more accurate but it increases computation time of system and also decreasing specificity parameter. Although fall detection systems which can send alert when fall event occurs are available with different technologies, Internet of the Things approach to notify person at remote side will be much better.

Additional improvement can be done as anesthesia detection, person identification and then tracking him only. If one can compromise the cost and complexity of system then multiple camera approach will be helpful to cover maximum area in room.

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REFERENCES

[1]. U. Nations, world prospects, 2008 revision, comprehensive tables: UN 2010.

[2]. M. Peden, "The Injury Chart Book".

[3]. Caroline Rougier, Jean Meunier, Alain St-Arnaud, and Jacqueline Rousseau, "Robust Video Surveillance for Fall Detection Based on Human Shape Deformation", IEEE transactions on circuits and systems for video technology, vol. 21, no. 5, may 2011.

[4]. Wild, D. Nayak, U.S. Isaacs, B., "How Dangerous are Falls in Old People at Home?", Br. Med. J.1981, 282, 266–268.

[5]. Frank Sposaro and Gary Tyson, "I-fall: an Android Application for Fall Monitoring and Response", 31st annual international conference of the IEEE EMBS Minneapolis, Minnesota, USA, September 2-6, 2009.

[6]. Khalil Niazmand, Claudius Jehle, Lorenzo T. D'Angelo and Tim C. Lueth, "A New Washable low-cost Garment for Everyday Fall Detection", 32nd annual international conference of the IEEE EMBS, Buenos Aires, Argentina, august 31 - September 4, 2010.

[7]. Won-Jae Yi, Oishee Sarkar, Sivisa Mathavan and Jafar Saniie, "Wearable Sensor Data Fusion for Remote Health Assessment and Fall Detection" , Electro/Information Technology (EIT), 2014 IEEE international conference.

[8]. Ronald Nocua, Norbert Noury, "Evaluation of Autonomic Nervous System for Fall Detection", sept.2009 IEEE EMBS.

[9]. Chansik Park, Jae_Wonsuh , " Pedestrian Navigation System with Fall Detection and Energy Expenditure Calculation", 2011 IEEE.

[10]. E. Auvinet, L. Reveret, A. St-Arnaud, J. Rousseau, and J. Meunier, "Fall Detection Using Multiple Cameras", Engineering in Medicine and Biology Society, 2008. EMBS 2008. 30th annual international conference of the IEEE, pages 2554–2557. IEEE, 2008.

[11]. R. Cucchiara, A. Prati, and R. Vezzani, "A Multi-Camera Vision System for Fall Detection and Alarm Generation", Expert Systems, 24(5):334–345, Nov2007.

[12]. N. Thome, S. Miguet, and S. Ambellouis, "A Real-time, Multiview Fall Detection System: A LHMM-based Approach", circuits and systems for video technology, IEEE transactions on, 18(11):1522–1532, 2008.

[13]. G. Diraco, A. Leone, and P. Siciliano, "An Active Vision System for Fall Detection and Posture Recognition in Elderly Healthcare", design automation & test in Europe conference & exhibition, pages 1536–1541, March 2010.

[14]. C. Rougier, E. Auvinet, J. Rousseau, M. Mignotte, and J. Meunier, "Fall Detection from Depth Map Video Sequences", International conference on smart homes and health telematics, pages 121–128, June 2011.

[15]. B. Mirmahboub, S. Samavi, N. Karimi, and S. Shirani, "Automatic Monocular System for Human Fall Detection Based on Variations in Silhouette Area", IEEE transactions on biomedical engineering, 60:427–436, Feb 2013.

[16]. C. Rougier, J. Meunier, A. St-Arnaud, and J. Rousseau, " Robust Video Surveillance for Fall Detection Based on Human Shape Deformation.", IEEE transactions on circuits and systems for video technology, 21:611–622, may 2011.

[17]. W. Feng, R. Liu, and M. Zhu, "Fall Detection for Elderly Person Care in A Vision-Based Home Surveillance Environment Using A Monocular Camera", signal, image and video processing, 8(6):1129–1138, 2014.

[18]. B. Toreyin, and Y. Dedeoglu, "HMM Based Falling Person Detection Using both Audio and Video" , 14th IEEE int. signal processing and communications applications,2006, pp.1-4.

[19]. G. Diraco, A. Leone, and P. Siciliano, "An Active Vision System for Fall Detection and Posture Recognition in Elderly Healthcare," in conf. design, automation & test in europe (date), pp. 1536-1541.

[20]. E. Auvient, F.Multon, A. Saint-Arnaud, J. Rousseau, and J.Meunier, "Fall Detection with Multiple Cameras: an Occlusion-Resistant Method Based on 3-d Silhouette Vertical Distribution," IEEE trans. information technology in biomedicine,vol. 15, pp. 290-299, 2011.

[21]. Zoran Zivkovic, Ferdinand van der Heijden." Efficient Adaptive Density Estimation Per Image Pixel for the Task of Background Subtraction", 2006

[22]. Zoran Zivkovic," Improved Adaptive Gaussian Mixture Model for Background Subtraction", ICPR 2004.

[23]. 2015. Image moment - Wikipedia, the free encyclopedia. [ONLINE] Available: http://en.wikipedia.org/wiki/ Image moment. [Accessed 26 April 2015].

[24]. M. K. Hu, "Visual pattern recognition by moment invariants", IRE Trans. Info. Theory, vol. IT-8, pp.179–187, 1962.

[25]. J. Flusser and T. Suk, "Rotation Moment Invariants for Recognition of Symmetric Objects", IEEE Trans. Image Proc., vol. 15, pp. 3784–3790, 2006.

[26]. M. K. Hu, "Visual Pattern Recognition by Moment Invariants", IRE Trans. Info. Theory, vol. IT-8, pp.179–187, 1962

[27]. Hu Moments OPEN CV Method [online]. Available: http: //docs.opencv.org/modules/imgproc/doc/structural_analysis_and_shape_descriptors.html?highlight=cvmatchshapes#humoments.

[28]. Python Sending Email using SMTP [online]. Available: https://www.tutorialspoint.com/python/python_sending_email.htm.

[29]. Base64 [online].Available: https://en.wikipedia.org/wiki/Base64.

[30]. MIME-Wikipedia [online]. Available: https://en.wikipedia.org/wiki/MIME#Content-Transfer-Encoding.

[31]. Motion-web home[online] .Available : http://www.lavrsen.dk/foswiki/bin/view/Motion/WebHome

[32]. Flask(web framework)[online].Available: https://en.wikipedia.org/wiki/Flask_(web_framework)